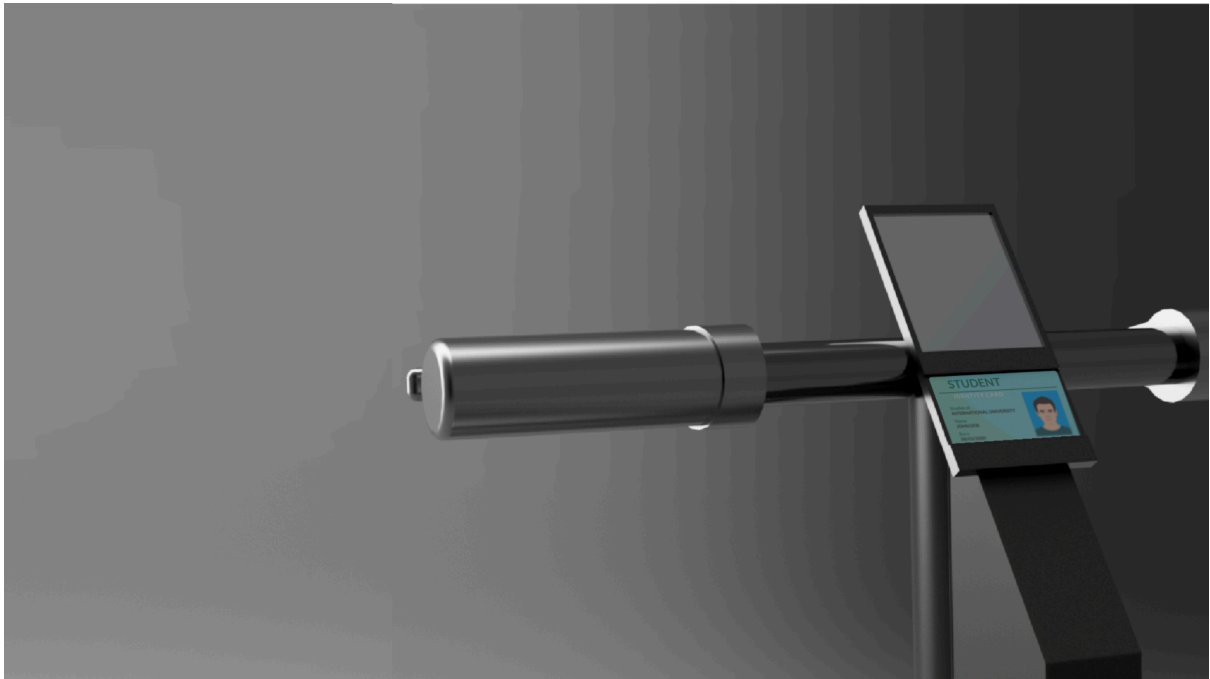


01 — HERO

CONNECT-CAMPUS

The on-site “last mile”.

A campus-specific micromobility system designed to make short-distance travel faster, more accessible, and easier to manage.



Industrial Design · Mobility · Systems Design

02 — BRIEF

Large campuses behave like small cities. Their mobility doesn't.

Connect-Campus explores an on-demand mobility system for campus environments, where short journeys are often constrained by walking distances, fixed shuttle schedules, and poorly integrated shared mobility.

03 — OVERVIEW

A mobility system, not just another scooter.

The concept combines a **smart vehicle, charging hub, and geo-lock system** into one controlled ecosystem.

The vehicle provides the ride, the hub manages where vehicles live and charge, and the geo-lock keeps the system within its intended operating environment.

04 — PRODUCT / SYSTEM

Three components keep the system moving.

SMART VEHICLE

A commercial-grade vehicle designed around frequent shared use.

CHARGING HUB

A dedicated parking and charging point that keeps the fleet organised.

GEO-LOCK

A digital control layer managing access, speed and operating zones.

05 — WHAT DO WE KNOW?

The problem was not a lack of mobility. It was a lack of fit.

I looked at existing public and private mobility options and combined that market study with **15 one-to-one interviews and a 250-person survey** to understand travel habits, frustrations and unmet needs.

06 — INSIGHT

Users needed flexibility without the chaos of public micromobility.

Public app-based fleets offered flexibility but created parking and pathway problems; private shuttles were more controlled but tied to fixed routes and schedules. The opportunity sat between them: **on-demand mobility with campus-level control.**

07 — DESIGN RESPONSE

The system had to control the journey, not just enable it.

I developed the vehicle together with the infrastructure around it, so access, charging, parking and location control became part of the product rather than separate services.

08 — FORM

The footprint became a design constraint.

Early concepts explored different vehicle architectures before narrowing toward a compact reverse-trike configuration suited to the spatial constraints of campus pathways.

09 — FIT

The right form was the one that could move through the campus.

The concepts were evaluated against footprint, pathway compliance and estimated cost. Concepts A and B were rejected, while the reverse-trike direction was selected for fitting the required path conditions.

10 — FUNCTION

Built for repeated use.

The selected vehicle combines a **6061 aluminium frame**, **swappable 48V battery**, **solid 10-inch tyres**, **onboard 4G IoT/GPS**, and **IP67 protection** to support fleet-oriented operation.

11 — EXPERIENCE

The ride becomes a closed loop.

A user requests access at a hub, the system authenticates and unlocks the vehicle, the ride is monitored through the connected system, and the vehicle returns to a hub where it is secured and recharged.

12 — DESIGN LANGUAGE

Robust where it needs to be. Quiet where it can be.

The vehicle uses a controlled structural frame with integrated technology, durable surfaces and environment-specific CMF rather than treating the finish as a purely aesthetic layer.

14 — USER / PROTOTYPE EVIDENCE

The system had to be designed beyond the vehicle.

The strongest evidence here is the interaction between the rider, hub and connected system: access, tracking, controlled movement and return all have to work together for the vehicle to function as shared infrastructure.

IMAGE

Use the **system-architecture / journey storyboard** as the primary visual.

For this project, I would **not claim physical prototype validation**, because the supplied case study does not document a completed physical test with users.

15 — BEAUTY SHOTS

CONNECT-CAMPUS

No explanation here.

IMAGE 01

Vehicle docked in hub

IMAGE 02

Vehicle / body detail

IMAGE 03

Handlebar + interface

These are already represented in the final-render section of your PDF.

Make these large and let the product breathe.
